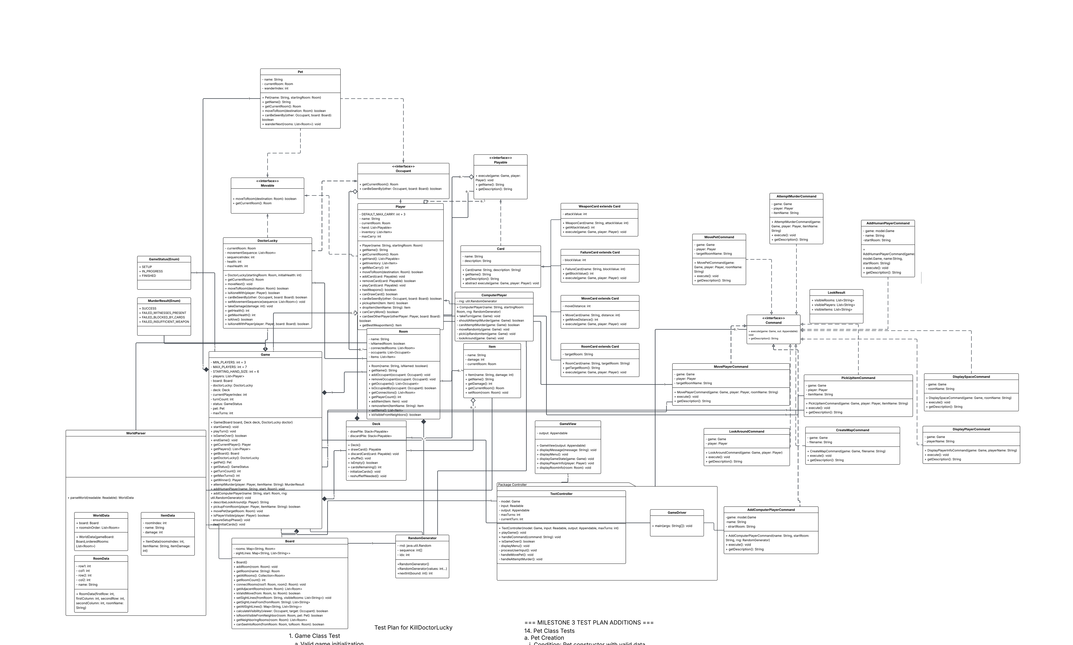

Test Plan for KillDoctorLucky

1. Game Class Test
    a. Valid game initialization
        i. Condition: Game constructor with valid player names
        ii. Example Data: ["Peter", "Will", "Alice", "Henry"]
        iii. Expected: the Game created, gameStatus = SETUP
    b. Minimum Players(Valid)
        i. Condition: Game constructor with minimum players to be 3
        ii. Example Data: ["Peter", "Will", "Alice"]
        iii. Expected: Game created successfully.
    c. Maximum Players (Valid)
        i. Condition: Game constructor with maximum players to be 7
        ii. Example Data: ["Peter", "Will", "Alice", "Henry", "Alan", "Olivia", "George"]
        iii. Expected: Game created successfully.
    d. Below Minimum Players
        i. Condition: Game constructor with 2 players
        ii. Example: ["Peter", "Henry"]
        iii. Expected: Exception thrown
    e. Above Maximum Players
        i. Condition:Game constructor with 8 players
        ii. Example:["Peter", "Will", "Alice", "Henry", "Alan", "Olivia", "George", "Jason"]
        iii. Expected: Exception thrown
    f. Start Game
        i. Condition: startGame called
        ii. Example: Game with 3 players
        iii. Expected: gameStatus changes from SETUP to IN_PROGRESS
    g. Successful Murder
        i. Condition: attemptMurder() with player alone, strong weapon
        ii. Example: Alice alone with Doctor Lucky, WeaponCard(value = 10)
        iii. Expected: MurderResult.SUCCESS, game over.
    h. Failed Murder - Witness
        i. Condition: attempMurder() with witness present
        ii. Example: Alice, Henry and Doctor Lucky all in same room
        iii. Expected: MurderResult.FAILED_WITNESS_PRESENT
2. Player Class Test
    a. Player Creation
        i. Condition: Player constructor with valid data
        ii. Example: name = "Alice", startingRoom = Kitchen
        iii. Expected: Player created, currentRoom = Kitchen, empty hand
    b. Valid Movement
        i. Condition: moveToRoom() to connected room
        ii. Example: Player in Kitchen, move to Dining Room
        iii. Expected: Returns true, currentRoom = Dining Room
    c. Invalid Movement
        i. Condition: moveToRoom() to unconnected room
        ii. Example: Player in Kitchen, move to Bedroom (not connected)
        iii. Expected: Return False, currentRoom unchanged
    d. Add Card to Hand
        i. Condition: addCard() with space available
        ii. Example: Player with 3 cards, WeaponCard("Knife", 5)
        iii. Expected: hand. size() = 4, card added
    e. Hands initialization
        i. Condition: Player starts with 6 cards
        ii. Example: New Game initialization
        iii. Expected: each player's hand.size() = 6
3. Board and Room Test
    a. Get Valid Room
        i. Condition: getRoom() with existing room name
        ii. Example: getRoom("Kitchen")
        iii. Expected: Returns Kitchen room object
    b. Invalid Room
        i. Condition: getRoom() with non-existent room
        ii. Example: getRoom("InvalidRoom")
        iii. Expected: Returns null or throws exception
    c. Add Occupant to Room
        i. Condition: addOccupant() with player
        ii. Example: add Henry to Kitchen
        iii. Expected: Henry in Kitchen's occupant list
    d. Room Visibility
        i. Condition: calculateVisibility() between rooms
        ii. Example: Alice in Kitchen and Bob in Dining Room (connected)
        iii. Expected: Returns True (can see each other)
4. Doctor Lucky Test
    a. Sequential Movement
        i. Condition: moveNext() called multiple times
        ii. Example: Movement sequence [Kitchen, Dining, Living], call 3 times
        iii. Expected: Kitchen to Dining to Living to Kitchen (circle)
    b. Alone Check-True
        i. Condition: isAloneWith() with only player present
        ii. Example: Kitchen has only Alice and Doctor Lucky
        iii. Expected: doctorLucky.isAloneWith() return True
    c. Alone Check-False
        i. Condition: isAloneWith() with multiple player
        ii. Example: Kitchen has Alice, Henry and Doctor Lucky
        iii. Expected: doctorLucky.isAloneWith() return False
5. Card Test
    a. Weapon Card
        i. Condition: WeaponCard creation and execution
        ii. Example: WeaponCard("Knife", 5)
        iii. Expected: Card created attackValue = 5
    b.Failure Card
        i. Condition: FailureCard blocks murder attempt
        ii. Example: FailureCard("Bad Luck", 3) used during murder
        iii. Expected: reduce murder success by 3 points
    c.Move Card
        i. Condition: MoveCard allows extra movement
        ii. Example: MoveCard("Quick Step", 2)
        iii. Expected: Player can move up to 2 additional rooms
    d. Room Card
        i. Condition: RoomCard instant teleportation
        ii. Example: RoomCard("Kitchen Card", "Kitchen")
        iii. Expected: player instantly move to Kitchen
6. Deck Test
    a. Empty Deck Reshuffle
        i. Condition: drawCard() when draw pile empty
        ii. Example: Empty draw pile, 20 cards in discard
        iii. Expected: discard reshuffled, card drawn
    b. Draw Card
        i. Condition: drawCard() with cards available
        ii. Example: Deck with 50 cards
        iii. Expected: return card, deck size decrease
    c Discard Card
        i. Condition: discardCard() adds to discard pile
        ii. Example: discard WeaponCard("Knife", 5)
        iii. Expected: Card added to discard pile
7. Interface Test
    a. Playable Interface
        i. Condition: All cards implement Playable
        ii. Example: WeaponCard, MoveCard treated as Playable
        iii. Expected: execute(), getName() work for all cards
    b. Movable Interface
        i. Condition: Player and Doctor Lucky implement Movable
        ii. Example: Both objects as Movable type
        iii. Expected: moveToRoom(), getCurrentRoom() work for both
8. Integration Test
    a. Game end
        i. Condition: Game ends when murder successful
        ii. Example: Alice successfully murders Doctor Lucky
        iii. Expected: gameStatus = FINISHED, Alice declared winner
    b. Multiple player interaction
        i. Condition: players interact during murder attempt
        ii. Example: Alice attempts murder, Bob plays FailureCard
        iii. Expected: Bob's card affects Alice's murder attempt
    c. Complete Turn
        i. Condition: full player turn with movement and card play
        ii. Example: Alice moves, plays MoveCard, attempts murder
        iii. Expected: all actions execute correctly, turn passes to next player
9.Controler Test
    a. Display Menu
        i. Condition: playGame() starts and displays menu
        ii. Example: TestController with Game
        iii. Expected: Menu displays all available commands
    b. Process Valid Command
        i. Condition: handleCommand() with valid input
        ii. Example: input "move Kitchen"
        iii. Expected: Player moves, success message displayed
    c. Process Invalid Command
        i. Condition: handleCommand() with invalid input
        ii. Example: input "xyz123"
        iii. Expected: Error message game continues
    d.MaxTurns Limit
        i. Condition: Game reaches maxTurns
        ii. Example: maxTurns = 10, play 10 turns
        iii. Expected: Game ends, final state displayed
    e.Game Over Detection
        i. Condition: isGameOver() returns true when done
        ii. Example: After max turns or successful murder
        iii. Expected: returns true, game stops
10. Command Pattern Test
    a. MovePlayerCommand Execution
        i. Condition: execute() moves player
        ii. Example: MovePlayerCommand(game, alice, "Kitchen")
        iii.Expected: alice.getCurrentRoom() = Kitchen
    b. PickUpItemCommand Execution
        i. Condition: execute() picks up Item
        ii. Example: PickUpItemCommand(game, alice, "Knife")
        iii.Expected: alice.getHand() contains Knife
    c. LookAroundCommand Execution
        i. Condition: execute() displays surroundings
        ii. Example: LookAroundCommand(game, alice)
        iii. Expected: Returns list of visible rooms/players
    d.Command Polymorphism
        i. Condition: All Commands work as Command interface
        ii. Example: Command cmd = new MovePlayerCommand(...)
        iii. Expected: cmd.execute() works correctly
11. ComputerPlayer Test
    a. Automatic Action Selection
        i. Condition: chooseAction() returns valid Command
        ii. Example: ComputerPlayer with RandomGenerator(1,2,3)
        iii. Expected: Returns MovePlayerCommand predictably
    b. Random Behavior
        i. Condition: Different RandomGenerator gives different actions
        ii. Example: Two ComputerPlayers with different seeds
        iii. Expected: Make different choices
    c. Valid Moves Only
        i. Condition: Computer never chooses invalid move
        ii. Example: ComputerPlayer in corner room
        iii. Expected: Only chooses valid neighboring rooms
12. View Test
    a. Display Message
        i. Condition: displayMessage() outputs to Appendable
        ii. Example: GameView with StringBuffer, displayMessage("Hello")
        iii. Expected: StringBuffer contains "Hello"
    b. Display Game State
        i. Condition: displayGameState() shows current info
        ii. Example: Game with 3 players, turn 5
        iii. Expected: Output shows player positions and turn count
13. RandomGenerator Test
    a. Predictable Mode
        i. Condition: Constructor with varargs returns predefined values
        ii. Example: RandomGenerator(1,2,3), call nextInt(10) three times
        iii. Expected: Returns 1, 2, 3 in order
    b. Random Mode
        i. Condition: Default constructor gives random values
        ii. Example: RandomGenerator(), call nextInt(10) multiple times
        iii. Expected: Returns different values each time
    c. Cycle Through Values
        i. Condition: Predefined values cycle when exhausted
        ii. Example: RandomGenerator(1,2), call 5 times
        iii. Expected: Returns 1,2,1,2,1

=== MILESTONE 3 TEST PLAN ADDITIONS ===

14. Pet Class Tests
    a. Pet Creation
        i. Condition: Pet constructor with valid data
        ii. Example: Pet("Fortune the Cat", kitchen)
        iii. Expected: Pet created, currentRoom = kitchen
    b. Pet Movement
        i. Condition: moveToRoom() to any room
        ii. Example: pet.moveToRoom(library)
        iii. Expected: Returns true, pet in library
    c. Pet Blocks Visibility
        i. Condition: Room with pet cannot be seen by neighbors
        ii. Example: Pet in kitchen, player in dining room
        iii. Expected: Kitchen not visible from dining room
    d. Pet Wandering (DFS)
        i. Condition: wander() follows DFS pattern
        ii. Example: Rooms in order [Kitchen, Dining, Library, Bedroom]
        iii. Expected: Pet visits in DFS order, cycles when complete
15. Murder Attempt Tests
    a. Successful Murder with Item
        i. Condition: Player alone with Doctor Lucky, uses weapon
        ii. Example: Alice alone, uses Knife (damage=5), health=50
        iii. Expected: MurderResult.SUCCESS if damage >= health
    b. Successful Poke in Eye
        i. Condition: Player alone with Doctor Lucky, no items
        ii. Example: Alice alone, no inventory, pokes eye (damage=1)
        iii. Expected: MurderResult.SUCCESS if health <= 1
    c. Failed Murder - Witness Present
        i. Condition: Another player can see the attempt
        ii. Example: Alice attempts murder, Bob in adjacent visible room
        iii. Expected: MurderResult.FAILED_WITNESS_PRESENT
    d. Failed Murder - Pet Blocks View
        i. Condition: Pet in room blocks neighbors from seeing
        ii. Example: Alice murders in kitchen, pet in kitchen, Bob in dining
        iii. Expected: MurderResult.SUCCESS (Bob can't see)
    e. Item Removed After Use
        i. Condition: Item used in murder attempt is removed
        ii. Example: Alice uses Knife, successful attempt
        iii. Expected: Knife removed from game, not in inventory or room
    f. Computer Player Attempts Murder
        i. Condition: Computer player attempts when alone and has best weapon
        ii. Example: ComputerPlayer alone with Doctor Lucky, has Revolver
        iii. Expected: Attempts murder with highest damage weapon
16. Pet Movement Command Tests
    a. Move Pet to Valid Room
        i. Condition: movePet() to existing room
        ii. Example: MovePetCommand("Library")
        iii. Expected: Pet moves to Library, counts as turn
    b. Move Pet to Invalid Room
        i. Condition: movePet() to non-existent room
        ii. Example: MovePetCommand("InvalidRoom")
        iii. Expected: Error message, turn not consumed
    c. Pet Movement Affects Visibility
        i. Condition: After pet moves, visibility changes
        ii. Example: Pet moves from Kitchen to Library
        iii. Expected: Kitchen visible again, Library now hidden
17. Enhanced Look Around Tests
    a. Look Around Shows Pet Location
        i. Condition: describeLookAround() when pet in room
        ii. Example: Player in Kitchen, pet in Kitchen
        iii. Expected: Description includes pet name and location
    b. Look Around Respects Pet Invisibility
        i. Condition: describeLookAround() when pet in neighbor
        ii. Example: Player in Kitchen, pet in Dining Room
        iii. Expected: Dining Room not in visible neighbors list
    c. Look Around Shows Doctor Lucky Location
        i. Condition: describeLookAround() shows target location
        ii. Example: Player in Kitchen, Doctor Lucky in Library
        iii. Expected: Description shows Doctor Lucky's location if visible
18. Game End Conditions Tests
    a. Game Ends - Successful Murder
        i. Condition: Player kills Doctor Lucky
        ii. Example: Alice reduces health to 0
        iii. Expected: gameStatus = FINISHED, winner = Alice
    b. Game Ends - Max Turns Reached
        i. Condition: turnCount reaches maxTurns
        ii. Example: maxTurns=20, play 20 turns without murder
        iii. Expected: gameStatus = FINISHED, winner = null
    c. Game Continues After Failed Murder
        i. Condition: Failed murder attempt doesn't end game
        ii. Example: Alice attempts murder, fails due to witness
        iii. Expected: gameStatus = IN_PROGRESS, play continues
19. Enhanced Player Information Tests
    a. Display Shows Target Location
        i. Condition: Current turn info includes Doctor Lucky location
        ii. Example: Start of turn display
        iii. Expected: Shows "Doctor Lucky is in: [room name]"
    b. Display Shows Pet Location
        i. Condition: Current turn info includes pet location
        ii. Example: Start of turn display
        iii. Expected: Shows "Fortune the Cat is in: [room name]"
    c. Display Shows Remaining Turns
        i. Condition: Turn counter shows progress
        ii. Example: Turn 5 of 20
        iii. Expected: Shows "Turn 5 / 20"
20. Computer Player Enhanced AI Tests
    a. Computer Chooses Murder When Possible
        i. Condition: Computer player alone with target, has weapon
        ii. Example: ComputerPlayer in same room as Doctor Lucky, alone
        iii. Expected: Calls attemptMurder() with best weapon
    b. Computer Uses Best Weapon
        i. Condition: Computer selects highest damage weapon
        ii. Example: Has Knife(5), Rope(3), Revolver(8)
        iii. Expected: Uses Revolver (damage=8)
    c. Computer Pokes When No Weapons
        i. Condition: Computer attempts murder without items
        ii. Example: ComputerPlayer alone, empty inventory
        iii. Expected: Attempts poke in eye (damage=1)
    d. Computer Moves Pet Strategically
        i. Condition: Computer may move pet to hide rooms
        ii. Example: Computer wants to hide pet in Kitchen
        iii. Expected: May move pet to Kitchen before entering
21. Integration Tests for Milestone 3
    a. Complete Game with Murder Victory
        i. Condition: Full game ending in successful murder
        ii. Example: 3 players, Alice wins on turn 15
        iii. Expected: Game ends, Alice declared winner, proper messages
    b. Complete Game with Timeout
        i. Condition: Full game reaching max turns
        ii. Example: 3 players, maxTurns=10, no successful murder
        iii. Expected: Game ends at turn 10, no winner, Doctor escapes
    c. Pet Wandering Through Game
        i. Condition: Pet moves every turn automatically
        ii. Example: Play 5 turns, observe pet location
        iii. Expected: Pet follows DFS pattern through rooms
    d. Multiple Murder Attempts
        i. Condition: Several failed attempts before success
        ii. Example: Turn 1: Bob fails (witness), Turn 5: Alice succeeds
        iii. Expected: Failed attempts don't end game, success ends game
    e. Human and Computer Players Mixed Game
        i. Condition: Game with both human and computer players
        ii. Example: 2 humans, 1 computer, play full game
        iii. Expected: Computer acts automatically, humans prompted for input
22. Visibility System Tests
    a. Pet Hides Room from Neighbors
        i. Condition: calculateVisibility() with pet present
        ii. Example: Pet in Kitchen, check from Dining Room
        iii. Expected: Kitchen not visible from Dining Room
    b. Pet Doesn't Hide from Same Room
        i. Condition: Player in same room as pet can see
        ii. Example: Alice and pet both in Kitchen
        iii. Expected: Alice can see Kitchen contents
    c. Multiple Players Witness Check
        i. Condition: isPlayerVisible() checks all players
        ii. Example: Alice attempts murder, Bob and Charlie in game
        iii. Expected: Checks visibility to both Bob and Charlie
23. Item Management in Murder Tests
    a. Item Used and Removed
        i. Condition: Successful murder removes item from game
        ii. Example: Alice uses Knife successfully
        iii. Expected: Knife not in Alice's inventory, not in any room
    b. Item Retained on Failed Murder
        i. Condition: Failed murder keeps item in inventory
        ii. Example: Alice attempts with Knife, witness present
        iii. Expected: Knife still in Alice's inventory
    c. Multiple Items Available
        i. Condition: Player has choice of weapons
        ii. Example: Has Knife(5), Rope(3), chooses Knife
        iii. Expected: Uses specified weapon, not automatic
24. Controller Integration for Milestone 3
    a. AttemptMurder Command Input
        i. Condition: User inputs "attack [itemName]"
        ii. Example: "attack Knife"
        iii. Expected: Calls attemptMurder with item, displays result
    b. MovePet Command Input
        i. Condition: User inputs "movepet [roomName]"
        ii. Example: "movepet Library"
        iii. Expected: Moves pet, displays confirmation, counts as turn
    c. Display Murder Result
        i. Condition: Murder attempt result shown to user
        ii. Example: Failed attempt due to witness
        iii. Expected: Clear message explaining why attempt failed
    d. Display Game Over
        i. Condition: Game end displays winner/outcome
        ii. Example: Alice wins by murder
        iii. Expected: Congratulations message, final game state
    e. Display Remaining Turns Warning
        i. Condition: Warn when approaching max turns
        ii. Example: Turn 18 of 20
        iii. Expected: Message like "Only 2 turns remaining!"